

The Crossroads of LLM and Traffic Control: A Study on Large Language Models in Adaptive Traffic Signal Control

Mohammad Movahedi^{ID} and Juyeong Choi^{ID}

Abstract—Recent advancements in Large Language Models (LLMs) have ushered in opportunities to craft agents that exhibit human-like cognitive abilities, notably reasoning and planning. Leveraging the vast knowledge bases and sophisticated understanding of interrelations inherent in LLMs, this study explores their application in the dynamic and complex realm of traffic management. Specifically, we focus on Adaptive Traffic Control Systems (ATCS) as a critical use case, where designing effective traffic controllers parallels the cognitive tasks of reasoning and planning. This paper introduces a novel LLM-based agent development framework to create two types of traffic controllers: a Zero-Shot Chain of Thought approach and a Generally Capable Agent (GCA) approach. The latter integrates new knowledge from environmental interactions to enhance reasoning and planning capabilities. We implement and compare these controllers within a simulated traffic flow scenario at a single intersection using the Simulation of Urban Mobility (SUMO) against conventional traffic control methods, including fixed-time, gap-based, and delay-based controllers. Our results demonstrate that GCA-based controllers notably outperform traditional systems, reducing halted vehicle numbers by 48.03% and increasing average vehicle speed by 25.29%. These controllers also exhibit superior flexibility, generating diverse phase patterns that adapt dynamically to changing traffic conditions. This study underscores the transformative potential of LLMs in traffic management, promising significant enhancements in efficiency, responsiveness, and versatility in urban traffic control, ultimately improving urban life by mitigating traffic congestion's economic and environmental impacts.

Index Terms—Adaptive control, generally capable agent, intelligent transportation systems (ITS), reasoning and planning, large language model (LLM).

I. INTRODUCTION

RECENT strides in the development of Large Language Models (LLMs) have revealed a new opportunity for their application in reasoning and planning tasks. These models have demonstrated capability far beyond simple text processing, suggesting their potential for developing more complex applications such as system control. Among these, urban traffic management stands out as a domain where LLMs can significantly impact, specifically by advancing traffic control systems. By leveraging the cognitive-like processes of LLMs, there is an opportunity to revolutionize how cities manage

traffic flow, enhance efficiency, and improve the overall quality of urban life.

Many recent studies consider the novel wave of LLMs as the early-stage Artificial General Intelligence (AGI) [1], [2], [3] LLMs, with their ability to generate contextually appropriate, human-like text based on provided prompts, demonstrate a broader understanding of knowledge domains than the typical narrow understanding akin to the early stages of AGI. Additionally, LLM models are developed based on the attention mechanism. The attention mechanism in Generative Pre-Trained Transformer (GPT) models, often known as self-attention or transformer attention, allows the model to weigh and consider different input parts when generating the output [4]. The available LLM models can mimic how an expert human agent understands by incorporating an attention mechanism. These models can solve a problem by giving more attention to the relevant and applicable solutions when generating a response. Additionally, using attention mechanisms in the multi-layered architecture of LLM models enables capturing complex and abstract relationships between inputs, including long-range dependencies, contributing to the generation of rich and diverse responses. Researchers have proposed multiple approaches to reasoning and planning with LLM models [5], [6], [7].

In the context of transportation systems, system control—like traffic signal controllers—embodies reasoning and planning challenges as often one can identify the explicit objectives (e.g., improving traffic fluidity or minimizing travel time) that can be achieved through a sequence of actions (e.g., developing phase plans). Many researchers have tried to develop a comprehensive strategy for this problem using Narrow AI (NAI) methods. The approach they adopt can profoundly affect the reasoning and planning capability of the adaptive traffic signal controllers (ATSC) and, eventually, its devised phase plan. Researchers in other disciplines may have encountered similar challenges and problem structures and developed strategies to address them. This convergence of challenges across domains paves the way for an opportunity to transfer knowledge from other disciplines to develop a new controller. Given that LLMs are developed using diverse data, it is conceivable that an LLM-based traffic controller would have the innate capacity to autonomously assimilate this vast cross-disciplinary knowledge to improve its reasoning and planning capabilities and eventually lead to more adaptive traffic control strategies.

Received 12 October 2023; revised 1 May 2024 and 10 October 2024; accepted 7 November 2024. Date of publication 26 November 2024; date of current version 4 February 2025. The Associate Editor for this article was L. Qi. (Corresponding author: Juyeong Choi.)

The authors are with the FAMU-FSU College of Engineering, Tallahassee, FL 32310 USA (e-mail: mmovahedi@fsu.edu; jchoi@eng.famu.fsu.edu).

Digital Object Identifier 10.1109/TITS.2024.3498735

1558-0016 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See <https://www.ieee.org/publications/rights/index.html> for more information.

The role of traffic signal controllers in developing an effective traffic management strategy is undeniably crucial as it controls the flow of vehicles, determines the fluidity of urban traffic, and ultimately markedly improves users' commutes on a day-to-day basis. In 2022, traffic congestion cost US drivers \$81 billion in wasted time, fuel, and collisions. A typical driver spends 51 hours and consumes the equivalent of \$869 in fuel in traffic congestion [8]. On average, about 23,308 vehicles pass through each intersection daily, spending more than 8.4% of their travel time waiting [9]. In addition to alleviating urban traffic congestion, finding an optimal traffic control strategy can significantly reduce fuel consumption and lower transportation-related emissions [10]. However, this is not an easy goal to achieve. The varying nature of traffic demand and traffic flow complexities necessitates an adaptive approach to traffic control systems that can comprehend these characteristics and alter the strategy to minimize the adverse impacts of its interventions. For instance, fluctuations in traffic volume and tidal traffic patterns can challenge the non-adaptive traffic signal controllers since the core characteristics of the traffic stream, such as traffic flow pattern and volume, significantly change [11].

Traffic signal controllers are widely categorized into three major categories: fixed time, actuated, and adaptive traffic signal controllers. Fixed time controllers set cycle lengths, phase sequences, and green splits based on historical traffic data and stay the same regardless of the time of day or the amount of traffic on the road. These controllers may, however, be adjusted to accommodate different traffic patterns at different times of the day (for example, afternoon peak hour versus morning peak hour) [12]. Actuated traffic controllers choose their plans based on real-time data from sensors from plans that were optimized offline and developed based on historical traffic data [13], [14], [15], [16]. On the other hand, adaptive traffic signal controllers adjust real-time signal timings based on current traffic conditions. They use sensors or cameras to continuously monitor the traffic situation at intersections and along roads, extract data, and adjust their strategies to achieve a specific goal based on that [10]. In the last few decades, the large-scale deployment of traffic sensors on intersections [17], [18], [19], [20] and the development of crowd-sourced vehicle probe databases [21], [22], [23], [24], [25], [26] paved the road for the development of more sophisticated ATSC that leveraged artificial intelligence (AI). Many ATSCs, developed based on NAI techniques, are specifically designed to handle traffic signals by analyzing data collected from those databases.

Recent advancements in generative AI have unveiled promising new research avenues that could significantly enhance the efficacy of adaptive controllers. LLM-based controllers can mirror human intelligence's ability to understand, learn, and apply knowledge across various tasks. While NAI-based ATSCs have been the cornerstone of many traffic signal control mechanisms due to their effectiveness in handling well-defined tasks, their scope is limited to their training data and knowledge base. They may struggle to adapt to unexpected or complex situations [27], [28]. In contrast, LLM's adaptability and capability for transfer learning make it a potentially powerful tool in traffic signal control.

Real-world traffic scenarios are multifaceted, often unpredictable, and incredibly challenging to model and control, with many variables and unforeseen circumstances. In other words, LLM-based controllers might open the door to exploring novel ideas and problem-solving strategies and even consider new data sources and variables left untouched by researchers when they developed NAI-based controllers.

This study explores the development of LLM-based reasoning and planning controllers for adaptive traffic signal management. We aim to assess whether LLMs, with their vast knowledge bases and attention-based learning capabilities, can effectively navigate the complexities of real-time, dynamic environments, such as managing an isolated intersection. To illustrate how these capabilities can be integrated, we adopt a bottom-up approach. Initially, in Section III, we provide a comprehensive explanation of the agent design (Figure 1-Step 1). Subsequently, Section IV shows how these agents are implemented to develop two distinct types of controllers (Figure 1-Step 2). Unlike traditional approaches that utilize narrow AI (NAI) and depend on predefined logic and decision-making processes, LLM-based agents leverage observation and interactive decision-making. They evaluate the current state—factors like vehicle count and queue lengths at intersections—to select actions aimed at optimizing objectives, such as reducing travel times or enhancing network throughput. These actions include transitioning to new traffic phases or extending the current ones.

Traffic flow characteristics depend on numerous factors; therefore, evaluating a timing strategy in real-world situations is often difficult. Evaluating the proposed approaches in hypothetical scenarios and benchmarks will enable researchers to compare other methods and strategies against the proposed LLM-based controllers. Like many preceding studies [29], [30], [31], [32], conventional signal control strategies, such as Webster fixed-time, gap-based, and delay-based controllers, will be used to have a touchstone to make comparison possible across different approaches (Figure 1-Step 3). The performance of the proposed controllers is then evaluated in Section V within a simulated traffic environment using performance measures such as vehicle waiting times and average speeds, that were selected based on previous research [11], [33] and guidelines [34]. This evaluation assesses the LLM-based controllers' effectiveness relative to conventional traffic signal controllers, focusing on adaptability to traffic patterns, fluidity of traffic flow, specific traffic signal performance metrics, and overall algorithm safety performance (Figure 1-Step 4). The results demonstrate the potential of LLM-based controllers in developing effective solutions when it comes to complex dynamics problems within the context of transportation engineering, offering a glimpse into the future of urban traffic management.

II. RELATED WORK

The use of NAI and, more recently, LLMs heralds a paradigm shift, promising more responsive and intelligent controllers. As this study delves deep into the possibility of using LLM-based reasoning and planning strategies in the context of adaptive traffic signal controls, examining their interplay and potential synergies in refining and redefining

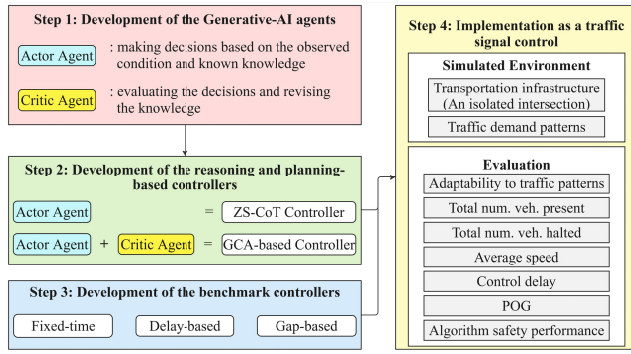


Fig. 1. The proposed framework for this study.

ATSC is necessary. This section presents an overview of existing literature on ATSC controllers and explores emerging research that employs LLMs for reasoning and planning.

In the context of adaptive traffic signal control, the LLM-based controllers exhibit remarkable similarities to reinforcement learning (RL) models, as both involve agents that evolve through continuous interaction with their environments. Similar to RL agents, which execute actions, receive rewards, and progressively refine their strategies, LLM-based controllers, particularly those using the GCA framework, monitor traffic dynamics, make decisions, and utilize feedback to enhance future performance. Both systems fundamentally rely on learning from past experiences to optimize outcomes. LLM-based controllers, including the GCA-based ones, actively update their knowledge and adjust their actions within a feedback loop, paralleling RL agents who adapt their policies based on received rewards.

A. Reasoning and Planning Using Large Language Models

The recent advancements in LLMs offer researchers a glimpse into the potential capabilities of forthcoming AGI on the horizon [1], [3], even though this glimpse might be an early-stage and partial representation of what an AGI model can do. Models like GPT-3.5 [35] and PaLM-2 [36] show remarkable capabilities across various domains and tasks. One specific field is reasoning and planning tasks and controlling feedback control systems. Since the introduction of LLM models, several applications in urban planning [37], agent-based simulation [38], [39], [40], and decision-making [41] have emerged.

Despite the vast capabilities in managing complex problems, the models may struggle with tasks humans find easy, such as generating plans for executing tasks in an environment or performing complex logical reasoning. On the other hand, these models have demonstrated impressive planning and reasoning capabilities when they are finetuned or prompted in a way that generates intermediate reasoning steps, such as with the Chain-of-Thought (CoT) approach [5]. To tackle this problem, several studies have proposed solutions to the limitations in reasoning and planning evident in LLMs.

Among these advancements, two techniques - Zero-shot Chain of Thought (Zero-shot-CoT) and Few-shot Chain of Thought (Few-shot-CoT) - have emerged as frontrunners, aiming to elevate the reasoning capabilities of LLMs. Zero-shot-CoT capitalizes on the model's inherent generalization

abilities, using the LLM's prior knowledge base for sequential reasoning without needing examples, showcasing a versatile aptitude for abstract and mathematic reasoning tasks [42]. In contrast, Few-shot-CoT optimizes the model's reasoning by introducing a select set of relevant examples, tuning its cognitive framework for improved solution accuracy and context alignment [43]. These techniques represent the confluence of intuition, generalization, and contextual guidance in modern LLMs. One can utilize an LLM-based model to conduct sophisticated reasoning and planning through a delicate chain of thoughts, step-by-step problem-solving explanations, or giving the LLM models examples of solved problems. When it comes to traffic signal control, the agent, the ATSC controller, must rationally plan and reason to find the best strategy for the intersection. Although applying the LLM models in reasoning and planning through agents is novel, a few studies have tackled this challenge.

Some researchers attributed the LLM's deficiency to their lack of an internal world model to predict the world state (e.g., environment status, intermediate variable values) and simulate long-term outcomes of actions [5]. This lack prevents LLMs from performing deliberate human-like planning, which involves exploring alternative reasoning paths, anticipating future states and rewards, and iteratively refining existing reasoning steps., principally by integrating them with process simulations. With the aid of LLMs, one can create Generally Capable Agents (GCAs) that dynamically engage with their learning environment. These agents can experiment with various planning techniques and reasoning methodologies to fulfill specific objectives. For instance, Hao [5] proposed Reasoning via Planning (RAP) as a framework to connect LLM models to learning environments with which to interact. RAP redefines the role of the LLM, utilizing it as both a world model and a reasoning agent, and incorporates a principled planning algorithm (based on Monte Carlo Tree Search) for strategic exploration in the vast reasoning space. In another study by Zhu et al. [44], integrating LLMs with text-based knowledge and memory was proposed as an advanced mechanism for agents to traverse intricate environments characterized by sparse rewards and predominantly text-based interactions. Zhu crafted a closed-loop LLM-driven controller specifically tailored to accomplish the "ObtainDiamond" task within Minecraft. This agent continuously assesses environmental feedback, refining its planning and reasoning processes based on feedback and a text-centric knowledge base. The agent leverages a structured set of actions and uses LLMs to design its plan carefully.

Furthermore, Wu et al. [45] proposed a framework that can provide the planning and reasoning capability for an agent playing a survival game and help it achieve its goals using knowledge gained through studying the technical documents and research papers on a domain and experimenting with the learning environment. Using Chain-of-Thought (CoT) prompts to leverage the GPT-4 model leads to outperforming the state-of-art RL models. Ahn et al. [46] took a significant step toward effective task planning by leveraging LLMs to premeditate an entire sequence of operations at the task's inception. This insightful approach eradicates the need for continual interaction with

the planner throughout the task, thereby increasing overall efficiency.

The Plan-and-Solve (PS) Prompting strategy has emerged as another innovative approach to address the limitations associated with multi-step reasoning using LLMs. Unlike Zero-shot-CoT prompting, which requires manual crafting of step-by-step reasoning demonstrations, PS Prompting focuses on systematically breaking down the entire task into smaller subtasks and executing them according to a pre-established plan [7]. This strategy has been designed to counteract missing-step errors, one of the three major pitfalls identified in existing Zero-shot-CoT studies.

In another research by Wang et al. [6] proposed a closed-loop feedback system called “Describe, Explain, Plan and Select” (DEPS), which interrupts task execution to seek a revised plan from the LLM when necessary, thus enhancing the robustness of the acting agent towards environmental uncertainties. However, this approach depends on hard-coded failure detectors, which can limit its general applicability. Concurrently, Dasgupta et al. [47] brought forth a novel Planner-Actor-Reporter framework aiming to improve the interaction between the actor and the LLM-based planner. In their model, multiple agents with different roles interact with the LLM at each step, irrespective of the emergence of new information. While this holistic approach helps better manage uncertainties and eliminates the reliance on hard-coded termination conditions, it is resource-intensive, particularly for larger and more costly LLMs.

LLMs have demonstrated remarkable zero-shot reasoning abilities, a phenomenon that is particularly evident when the models are provided with effective prompts. Approaches like zero-shot-CoT, which involves forming a well-structured prompting strategy, capitalize on this ability. These approaches can reason and plan within uncertain environments without prior explicit task-specific training. Zero-shot-CoT showcases the adaptability and versatility of LLMs in deciphering complex situations, suggesting that appropriately tailored prompting can significantly enhance the applicability and efficacy of zero-shot learning techniques [48].

Agents powered by LLM can, in theory, conduct reasoning and planning due to their intricate cognitive task performance, closely mirroring human reasoning and problem-solving strategies [42], [43] LLM-based agents embody knowledge transfer, mirroring transfer learning by utilizing their vast training to address new, complex challenges. Their capability to use context to infer meanings and grasp linguistic subtleties enables a human-like problem-solving approach, breaking tasks into manageable segments. Coupled with the LLMs’ extensive knowledge, this approach to reasoning and planning enables researchers to develop generally capable agents that can perform complex tasks.

Additionally, the LLMs are pre-trained models that were trained on a vast knowledge base, which allows for their flexible application across diverse domains without additional training, showcasing their adaptability in dynamic environments such as traffic management. Thus, LLMs highlight their crucial role in pioneering solutions for unforeseen problems, emphasizing their utility in adaptive decision-making [49]. Through the decomposition of questions and analysis of

real-time data, LLMs can generate reasoned, contextual responses and actionable insights, marking a significant advancement in AI’s ability to handle complex, dynamic decision-making scenarios.

B. Reinforcement Learning-Based Traffic Signal Controllers

In the early 1960s, Miller [50] introduced the concept of ATSC. His proposed strategy considers the changes in traffic demand and adjusts the traffic signal timing based on that. His research laid the groundwork for future research, many of them involving leveraging NAI methods, from fuzzy logic [51], [52], [53] to deep reinforcement learning (RL) [33], [54], to optimize traffic signal control strategies.

Over the past two decades, there has been substantial progress in RL-based adaptive traffic signal control development. Early work in this field focused predominantly on tabular Q-learning, which typically relied on a discrete state space [55]. For instance, when used as a state, queue length was often categorized into low, medium, and high levels based on predefined rules [56]. However, this simplistic representation may not adequately capture the traffic flow complexity, leading to suboptimal signal control [57].

The emergence of neural networks as a powerful tool for function approximation has significantly advanced the application of RL in traffic signal control. Studies like Li et al. [58] and Guo et al. [11] demonstrated how deep reinforcement learning can predict the long-term consequences of decisions and eventually help agents learn optimal control policies at a single intersection. Shabestary et al. [59] and Zeng et al. [60] extended this further, employing deep neural networks and memory-based RL methods, respectively, to optimize traffic signal control [59], [60] Furthermore, Liang proposed a novel approach that used a convolutional neural network (CNN) to match the states and expected future rewards [57].

A common theme in these studies is how they tried to achieve the optimal policy by employing various variables in the reward signal the agent receives and sophisticated methods it uses to predict the value of its actions. Most of the time, the relationship between the observed variables (e.g., travel time, waiting time, etc.) and the value of an action is complex. Some studies have focused on addressing this issue by integrating RL with deep learning to enhance RL’s capability to learn complex non-linear problems [10], [57], [58], [61], [62] Considering the recent prominent development, LLM models can help the controllers with reasoning and planning. An LLM-based adaptive traffic control system could draw knowledge from multiple domains, respond more effectively to real-time changes, anticipate future conditions, and make decisions that balance the needs of all road users, ultimately leading to more efficient, safer, and equitable traffic systems.

RL-based methodologies have profoundly influenced the evolution of adaptive traffic signal control (ATSC). Nevertheless, research into applying LLM-based controllers for ATSC is largely uncharted territory, creating a gap in the literature. Key challenges facing this line of research include the limitations of LLM-based reasoning and planning algorithms in generating executable action plans and performing complex mathematical and logical reasoning tasks and their absence

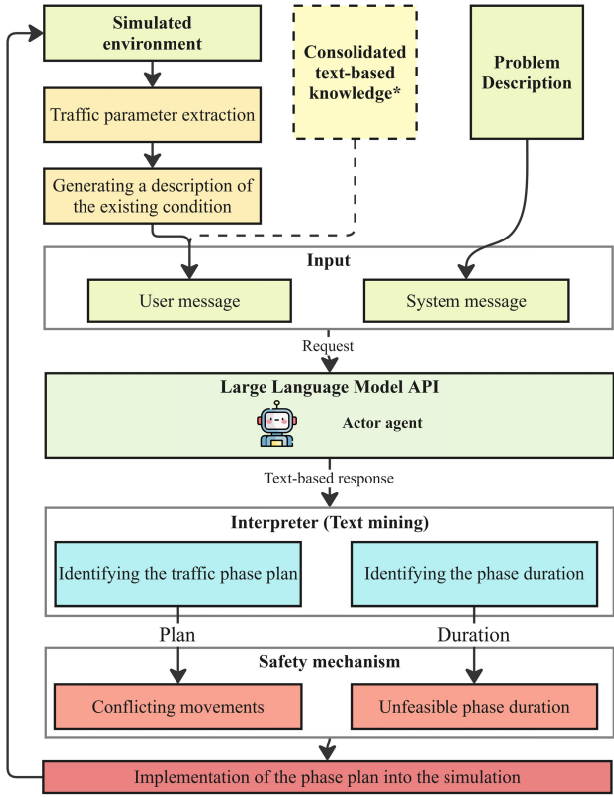


Fig. 2. The implementation of the LLM-based actor agent.

of an internal world model for predicting future states and long-term outcomes. These challenges can be addressed by integrating the LLM-based traffic signal controller that controls a single intersection with a microscopic traffic simulation.

III. LLM-BASED REASONING AND PLANNING AGENTS

In order to explain the developed framework, this study adopts a bottom-up perspective, beginning with a detailed examination of the agents involved before delving into the integration process to form a comprehensive controller. In this section, we initially focus on two types of Generally Capable Agents (GCAs), exploring the functions and roles of two primary types: the actor agent and the critic agent. The actor agent is responsible for making real-time decisions and taking actions within the system, while the critic agent evaluates the outcomes of these actions and provides feedback, enhancing the system's overall learning and performance.

A. Actor Agent

The actor agent represents the proactive component of the forward-feed flow in the decision-making architecture. This agent leverages three primary information sources to inform its decisions (Figure 2). Firstly, it utilizes a general description of the problem and the specific task it is meant to accomplish, such as generating traffic signal plans; this general problem description will remain constant and is stored as a system message. Secondly, it gathers real-time state parameters from the simulation environment to accurately assess current conditions. Exclusively integrated into the GCA-based controller design, the agent may have access to consolidated text-based

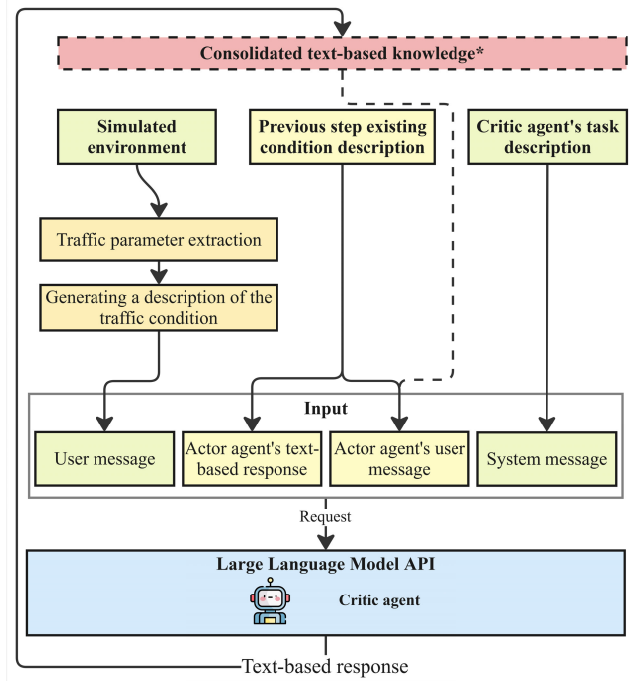


Fig. 3. The critic agent extracts and consolidates the text-based knowledge from the actor agent's interaction with the learning environment.

knowledge. The consolidated text-based knowledge is only implemented in one controller design (GCA-based controller) where there is a second agent, the critic agent, to generate text-based knowledge based on actor agent interactions with the learning environment.

As illustrated in Figure 2, using the generated description of the existing state and the consolidated knowledge, the agent will generate a user message that is later fed to the LLM model. The LLM model will generate a response that a text-mining algorithm interprets and extracts the recommended action (the recommended traffic phase and duration). Before implementing the generated action, a safety mechanism checks if the devised plan (i) does not allow two or more conflicting movements at the same time and (ii) does not violate the predefined acceptable phase duration range.

B. Critic Agent

As depicted in Figure 3, the actor agent's interactions can be evaluated by the critic agent. Like the actor agent, the critic agent uses a post-interaction description of the current traffic condition and integrates it with the actor agent's interaction with the environment. This process is designed to consolidate lessons learned from past actions by analyzing the outcomes of traffic simulations. With each cycle, the critic agent updates the text-based knowledge, providing the actor agent with a better understanding of the effective decision-making strategies for its next decision.

IV. LLM-BASED REASONING AND PLANNING AGENTS AS SIGNAL CONTROLLERS

ATSC can be systematically engineered by integrating these two types of agents. Therefore, one can use LLMs in two distinct approaches to enhance decision-making processes.

TABLE I
AN EXAMPLE OF THE INTERACTION STRUCTURE DEVELOPED FOR THE ZS-CoT CONTROLLER

Level	Message
System	Act as an adaptive traffic signal controller The environment: an intersection that connects four roads: north, east, south, and west. Each road has three lanes; the cars on the right lane can go through and turn right, the cars on the middle lane can only go through, and the vehicles on the left lane must turn left. Separate lights control the right and left turn signals. Task: In each query, you will receive a description of the traffic conditions at this intersection. Follow the following steps to find the optimal traffic light schedule in the given environment that reduces the waiting time at the intersection. Limitations: 1) The minimum phase duration is 30 seconds. 2) The maximum phase duration is 120 seconds. Steps: 1) Step 1: Decide how many roads will get the green light. 2) Step 2: Decide which road or roads will get the green light. 3) Step 3: Decide which roads are allowed to have right-turn green lights. 4) Step 4: Decide how long this phase should last based on the observed traffic condition. 5) Step 5: Summarize the output using the following format. a) The number of roads that get the green light. b) Roads that granted the green light c) list of the roads with right-turn green lights d) An integer that shows phase duration in seconds
User	1) North road a) Vehicle present: left-turn lane: 0 cars, middle lane: 1 car, and right-turn lane: 2 cars b) Queue length: left-turn lane: 0 cars, middle lane: 1 car, and right-turn lane: 0 car c) Average waiting time: 5 seconds 2) East road: a) Vehicle present: left-turn lane: 2 cars, middle lane: 1 car, and right-turn lane: 5 cars b) Queue length: left-turn lane: 1 car, middle lane: 0 cars, and right-turn lane: 5 cars c) Average waiting time: 65 seconds 3) South road: a) Vehicle present: left-turn lane: 0 cars, middle lane: 0 cars, and right-turn lane: 0 cars b) Queue length: left-turn lane: 0 cars, middle lane: 0 cars, and right-turn lane: 0 cars c) Average waiting time: 0 seconds 4) West road: a) Vehicle present: left-turn lane: 5 cars, middle lane: 2 cars, and right-turn lane: 3 cars b) Queue length: left-turn lane: 5 cars, middle lane: 2 cars, and right-turn lane: 1 car c) Average waiting time: 30 seconds

The first approach positions the LLM as an ‘actor agent,’ leveraging its extensive knowledge base and both advanced reasoning and planning capabilities to comprehend complex problems and orchestrate a sequence of actions. Here, only the actor agent engages in planning, analyzing variables, and selecting the most appropriate course of action in the given condition.

This Zero-Shot controller relies solely on the capabilities of the actor agent to operate without prior specific training on tasks, which resembles an open-loop feedback control system, where actions are determined based on a static knowledge base without feedback from the environment. Such an approach capitalizes on the LLM’s extensive training but remains imperious to real-time feedback or adaptation.

The second approach is based on a dual-agent framework that is engineered to simulate the systematic processes of human understanding, facilitating the processing of uncertain information from the surrounding environment based on previous studies on how the human understanding process works [63]. The framework operates similarly to a closed-loop feedback control system, consisting of two pathways: a forward-feed mechanism, where the agent selects actions based on its current information and knowledge; and a backward-feed mechanism, where the agent learns from its observations and refines its knowledge accordingly. The GCA-based controller incorporates both the actor and critic agents, harnessing their combined strengths to enable a more

adaptive control system capable of both decision-making and self-improvement through iterative feedback.

A. Zero-Shot Chain of Thought Controller (ZS-CoT)

The pipeline for implementing the ZS-CoT controller comprises four essential stages. Initially, a system message outlines the intersection’s general description, the controller’s objectives, and constraints. Subsequently, real-time traffic data is collected from the learning environment and relayed to the actor agent as a user message. Based on these inputs, the actor agent crafts a traffic signal strategy, which is then evaluated against predefined constraints to ensure its safety and feasibility. This structured approach provides an adaptive and data-informed traffic management solution.

Figure 4 illustrates the proposed ZS-CoT controller and its interaction with the learning environment. The system message outlines the geometry of the intersection (name of the streets, number of lanes, and allowable movement in each lane), a description of a user message structure, limitations of the designed phase (maximum and minimum duration), the primary goal of minimizing waiting time, and finally the expected format of the response is described.

Including the limitations is a critical part of this message, as it ensures the model does not produce impractical solutions that could compromise the safety of the users (e.g., overly long phase durations or unsafe phases with too many conflicting movements). In addition to a general description of

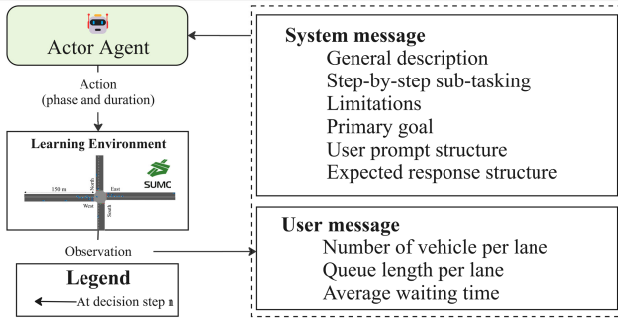


Fig. 4. The proposed ZS-CoT controller framework.

the problem, the system message provides the actor agent with a step-by-step subtask. The system message remains consistent throughout the simulation. Through the user message, the ZS-CoT controller can realize the state of the given intersection.

In the second stage, the user message includes a description of the average waiting time for each road, the number of vehicles present, and the queue length in each lane, which acts as the input for the controller. An example of the actor agent's message structure in this study is presented in TABLE I. Through the user message, the ZS-CoT controller can realize the state of the given intersection. The number of vehicles present and the queue length in each lane are used as the state realization, which is included in the user message.

In the third stage, the actor agent uses the LLM model, which is tasked with determining the optimal traffic control strategy based on one user and one system message provided. The actor agent checks the generated action based on the safety standards (e.g., conflicting movements) and feasibility criteria (e.g., minimum and maximum green time) to ensure the plan optimizes traffic flow and prioritizes users' safety.

B. Dual-Agent Controller (GCA-Based)

In contrast to the ZS-CoT approach, an actor-critic paradigm offers a more dynamic integration. Using this dual-agent concept, one can create a close-loop controller that utilizes a feedback loop to provide continuous insight into the consequences of previous actions, thereby offering a framework capable of iterative refinement.

Researchers have explored the concept of using text-based knowledge gained through interaction with the environment before [44], [45], [64], [65]. At the next decision point, the controller starts by updating its knowledge by initiating the backward path using the critic agent. The critic agent will receive the interaction that the actor agent had at the last decision point, including the actor's system message, user message, the API's response, as well as feedback from the learning environment at the current time step, quantifying the quality of the last decision the actor agent made. Based on this information, the critic agent will update the text-based knowledge, which the GCA-based controller forward path, the actor agent, will use to make a new decision. The actor agent in this framework is similar to the ZS-CoT controller but with text-based knowledge added to its system message. This part of the system message is the only part that will be updated at each decision point.

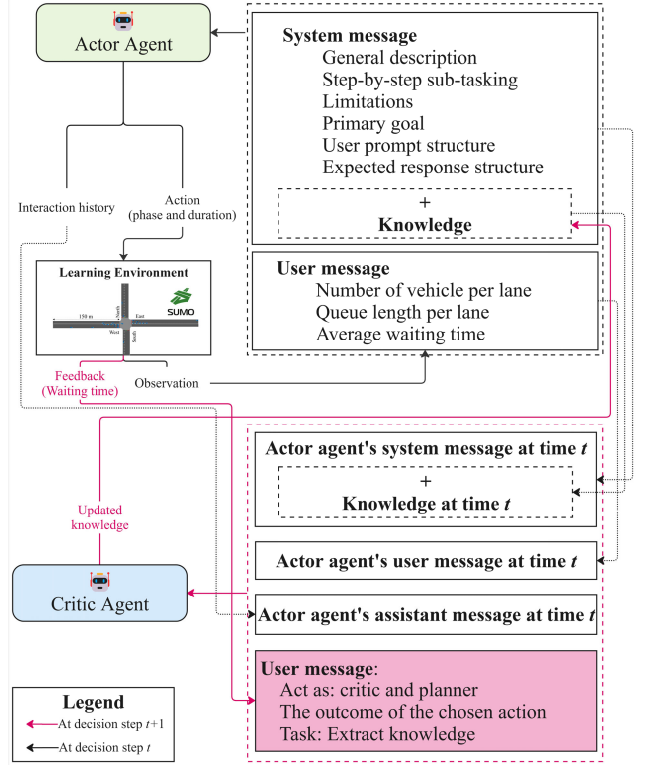


Fig. 5. The proposed GCA-based controller framework.

Figure 5 illustrates how the critic agent will receive the feedback, consisting of the waiting time of the vehicles, and update the knowledge to enable the critic agent to evaluate the selected strategy at the previous decision point and update the knowledge. Like the ZS-CoT controller, the actor agent's response in the GCA-based controller is checked to prevent unfeasible or unsafe phases from being implemented in the environment.

V. RESULT AND DISCUSSION

To assess the efficiency of the proposed frameworks, we structured an experimental evaluation using learning environments developed within the Simulation of Urban Mobility (SUMO) [66] and controlled by the Traffic Control Interface (TraCI) [67]. These environments simulate a diverse range of demand scenarios, each designed to test the performance and adaptability of the controllers under various challenging traffic conditions. To establish a benchmark for performance comparison, we also developed multiple conventional traffic controllers within the simulation. This experimental setup, with its array of scenarios and controllers, provides a testbed for evaluating the capabilities of the proposed traffic management solutions against established standards. In this study, rather than fine-tuning LLM agents to generate structured responses, we tried to harness them purely for their reasoning and planning prowess. The LLM model used in this study is OpenAI GPT 3.5-Turbo [68]. The model's temperature was set to zero to avoid unnecessary randomness in the responses.

A. Benchmark Traffic Signal Controllers

ATSCs have gained prominence in the domain of traffic management. Nevertheless, with an evolving landscape of

TABLE II
BENCHMARK TRAFFIC SIGNAL CONTROLLERS' PARAMETERS

Parameters	Value	Controller
Saturation flow rate	1900 veh/h/ln	FTSC
Yellow time	3 sec	All
All-red interval	1 sec	All
Minimum duration	30 sec	All
Maximum duration	120 sec	All
Maximum time gap between successive vehicles	5 sec	Gap-based
Minimum time loss of a vehicle	1 sec	Delay-based

methodologies, direct comparisons between emerging techniques and their predecessors present challenges. In the context of adaptive traffic signal controllers, establishing robust benchmarks serves as a foundation for methodological comparison. Through these benchmarks, it is possible to evaluate the performance of new traffic control strategies and compare them to other strategies proposed by other researchers. Recognizing this challenge, this study employs three benchmark controllers: the Webster fixed-time, gap-based, and delay-based models. These controllers, especially the Webster fixed-time traffic signal controller, have been widely used as benchmarks in previous studies [11], [18], [30], [31], [32], [56], [60], [61], [69], [70], facilitating subsequent comparative analysis and further development in future studies.

1) *Fixed-Time Signal Control*: Traditionally, the Fixed-time Signal Controllers (FTSC), particularly the Webster method, are used to facilitate the comparison between different signal control strategies. The Webster method is a pioneering technique that optimizes signal timings [71]. Its inception was driven by the ambition to elevate the efficiency and fluidity of vehicular movement at crucial intersection points.

2) *Gap-Based Signal Control*: Gap-based traffic signal control emerges as a pivotal mechanism based on the real-time dynamics of vehicular flow [69]. At its core, the Gap-based control scheme is geared towards extending traffic phases in instances where uninterrupted vehicular streams are discerned. A transition to the subsequent phase is executed once the system detects an adequate temporal gap between consecutive vehicles.

3) *Delay-Based Signal Control*: The delay-based traffic signal control emerges as a sophisticated variation of the conventional actuated control paradigm a gap-based traffic signal controller, this controller prolongs the active phase based on users' time loss. Delay-based controllers primarily monitor vehicular delay times and promptly adapt the light phase upon detecting a vehicle nearing a threshold delay time.

The parameters that are used to develop the benchmark traffic signal controllers are specified in TABLE II.

B. Scenario Development

The agents can learn how to make effective decisions by interacting with a learning environment that mirrors the intricacies of actual traffic conditions. To develop such an environment, a single four-legged signalized intersection is developed based on previous studies (See [11], [31]). Four distinct traffic patterns were generated to thoroughly evaluate the performance of a given traffic controller in different traffic demand settings, as described in TABLE III.

TABLE III
TRAFFIC DEMAND PARAMETERS

Pattern	Direction	Arrival rate (veh/sec)		
		←	↑↑	→
P1: Major/minor road	N-S & S-N	0.025	0.05	0.01
	E-W & W-E	0.05	0.10	0.01
P2: Through and left turn	N-S & S-N	0.025	0.05	0.01
	E-W & W-E	0.10	0.05	0.01
P3: Tidal traffic	N-S & E-W	0.08	0.10	0.01
	S-N & W-E	0.025	0.05	0.01
P4: Varying demand	N-S & S-N	0.025	0.05	0.01
	E-W	Steps 0-1200	0.05	0.01
		Steps 1200-1800	0.15	
	W-E	Steps 0-600	0.15	0.01
		Steps 600-1800	0.05	

The first pattern recreates a scenario where a minor road (North-South) intersects with a major road (East-West). The challenge here lies in optimizing traffic flow on the major road without overly delaying the minor road traffic. The second pattern reflects a situation where there is a high demand for left turns. Managing such high-demand left-turn scenarios can be tricky, requiring careful synchronization to avoid congestion and ensure safety. One way that the controller can handle this situation is to implement dedicated left-turn signals (protected phases) or allow left turns on a permissive basis when safe. The third pattern is designed to mimic tidal traffic conditions, where the demand from the North and East approaches significantly exceeds the other directions. In any four-legged intersection, each approach typically has the most conflicting movements with its adjacent approaches.

This presents a significant challenge for traffic controllers, especially in scenarios where the two major approaches are directly neighboring each other. This pattern often occurs during peak hours. Finally, the fourth pattern represents fluctuating traffic demand, where traffic volumes and directions vary dynamically. This complex pattern requires the controller to adapt to changing conditions to maintain optimal traffic flow constantly. Handling fluctuating traffic patterns is crucial for modern traffic management systems, especially in urban areas where traffic conditions can change rapidly due to various factors like events, weather conditions, and typical daily fluctuations. Each of these patterns tests the capabilities of traffic signal controllers, pushing the need for more intelligent systems that can adapt to complex and changing traffic conditions.

The simulation parameters were calibrated based on previous studies to reflect realistic traffic conditions [11], [31]. Each vehicle within the simulation is standardized with a length of 5 meters. For the car-following behavior, the Krauss following model was used with a minimum gap of 2.5 meters between each vehicle [72]. The vehicles are also constrained by a maximum speed limit of 13.42 meters per second, which aligns with urban traffic conditions. The maximum acceleration of vehicles is 2.6 m/s², and the maximum deceleration rate is 4.5 m/s². These parameters are integral to the simulation, as they govern the dynamics of vehicular movement and influence the overall traffic flow and safety within the simulated environment.

C. Quantitative Comparison of Performance Measures

By evaluating performance under different scenarios, it is possible to gain a robust understanding of the strengths and

limitations of the LLM approach in the development of adaptive traffic management systems.

The two-tailed Wilcoxon signed-rank test is employed to assess the performance differences among various controllers. This test is particularly invaluable when the data under examination violates the assumptions of parametric tests, such as the t-test. Specifically, it does not necessitate the data to adhere to a normal distribution. Within the traffic engineering domain, comparing different signal controllers' performance often results in datasets that may not satisfy the strict normality criteria, especially given the myriad of variables affecting traffic flow and control [73].

The primary goal of employing this test is to ascertain whether the performance measures for intersections managed by the proposed LLM-based controllers are significantly different from those controlled by conventional controllers. Therefore, the null hypothesis for this analysis is: "The implementation of LLM-based models does not have a significant impact on the performance measures when compared to conventional controllers." If the p-value is less than 0.05, the difference between the two controllers is considered statistically significant, implying that the LLM-based controller has a significant impact (positive or negative) on the performance measures. Based on the previous studies [11], [33] and guidelines [34], the performance of the developed controllers is evaluated in four performance measure categories:

1) *Adaptability to Traffic Patterns*: These performance measures evaluate the controller's ability to adapt to traffic patterns by generating new signal phase patterns and adjusting phase durations. While the conventional signal controllers are bound to their predefined phases, some ATSCs strategize their actions and generate more diverse phases in response to observed demand patterns. These patterns encompass situations where several movements are accommodated simultaneously. On the other hand, it is imperative to achieve a delicate balance between supplying all the demand and the effective use of green time. While excessive phase modifications could reduce the available green time, making it counterproductive, too infrequent adjustments could limit the controller's ability to respond to changing traffic dynamics in real time, culminating in diminished performance.

The green time-to-cycle (G/C) ratio is calculated by dividing the green phase duration by the total cycle length. This ratio measures the controller's efficiency in utilizing cycle time by showing the ratio of the cycle allocated to the green phase, thereby optimizing traffic flow at intersections.

TABLE IV enumerates the distinct phases generated by the traffic signal controllers, the phase change frequency, and the green time-to-cycle ratio. The data suggests that LLM-based controllers have tried to maintain this balance, avoided excessive phase changes, yet ensured they remain responsive to evolving traffic patterns by creating phases that minimize the traffic flow interruption. Both LLM-based controllers exhibited notable efficiencies when it came to the allocation of green time and capacity. Compared to the fixed-time controller, the ZS-CoT method allocated, on average, 11.23% more cycle time to the green time, raising the green-to-cycle ratio from an initial 68.68% to 79.88%. The GCA-based approach improved the green time allocation by 8.22%, from

TABLE IV
NUMBER OF DISTINCT PHASES OBSERVED (NP), THE FREQUENCY THAT THE CONTROLLER CHANGES THE PHASE, AND THE GREEN TO CYCLE RATIO (G/C) FOR EACH CONTROLLER-DEMAND COMBINATION IN PERCENTAGE

Controller	PM	P1	P2	P3	P4	Average
FTSC	NP	4	4	4	4	4
	Freq	0.096	0.096	0.083	0.083	0.09
	G/C	66.40	66.40	70.95	70.95	68.68
Gap-based	NP	4	4	4	4	4
	Freq.	0.078	0.064	0.021	0.089	0.063
	G/C	72.70	77.60	92.65	68.85	77.95
Delay-based	NP	4	4	4	4	4
	Freq.	0.088	0.071	0.037	0.093	0.072
	G/C	69.20	75.15	87.05	67.45	74.71
ZS-CoT	NP	14	18	15	11	14.5
	Freq.	0.06	0.061	0.057	0.052	0.058
	G/C	79.00	78.65	80.05	81.80	79.88
GCA-based	NP	11	12	19	6	12
	Freq.	0.071	0.068	0.068	0.057	0.066
	G/C	75.15	76.20	76.20	80.05	76.90

68.68% to 76.90%. These enhancements were achieved by minimizing unnecessary phase changes and redirecting the conserved time toward passing traffic, ultimately increasing the movements' capacities [12]. In traffic patterns 1 and 4, characterized by fewer conflicting movements, the gap-based and delay-based controllers exhibit a lower green time-to-cycle (G/C) ratio compared to LLM-based controllers. In contrast, these controllers had higher G/C ratios in pattern 3 and pattern 2, where there is a high demand for conflicting movements. The functionality of gap-based and delay-based controllers, which rely on delays and gaps between vehicles, means that in situations with frequent conflicts, the criteria for changing phases are triggered less frequently. This leads to a significant reduction in the frequency of phase changes. Moreover, this dynamic is also reflected in the number of phase patterns generated. The LLM-based controllers could effectively manage these conflicting movements by generating more tailored phases.

2) *Traffic-Related Performance Measures*: These performance measures are focused on the traffic flow characteristics and how the user's experience will change when implementing LLM-based controllers. Traffic-related performance measures are evaluated at the intersection and approach level to better understand the controller's chosen strategy. Evaluating performance at the intersection level provides a holistic view of the efficiency of the traffic control system. Meanwhile, approach-level assessment offers a more detailed, microscopic view and how the controller operates.

a) *Total number of vehicles present in the network (NVP)*: The total number of vehicles present in the network at the same time serves as an indicator of a controller's continuous performance. Considering this study's demand pattern, an ideal controller should manage traffic demand to minimize this count by increasing the network's throughput.

The throughput of a traffic network in a time interval can be determined by analyzing the number of vehicles entering or departing the network at the beginning and end of that interval and the vehicles that could not enter the network due to congestion during this period [74]. Therefore, the total number of vehicles present in the network at the same time could be an effective measure to compare the continuous performance of

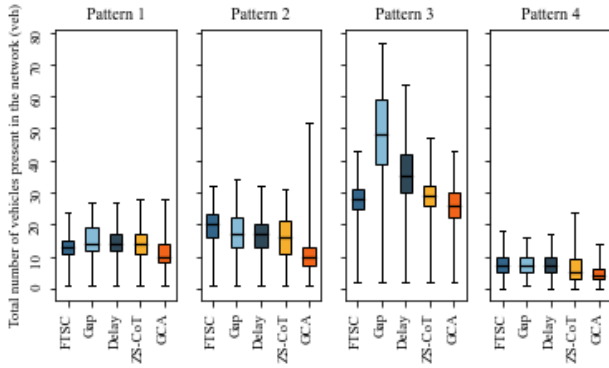


Fig. 6. Total number of vehicles present in the network at the intersection level.

controllers, influenced not only by the controller's immediate actions but also by all preceding steps.

As shown in Figure 6, the LLM-based controllers can manage to have fewer vehicles present in the network at the same time, demonstrating their strategic continuous planning capabilities. The Wilcoxon signed-rank test results presented in TABLE IV confirm that implementing ZS-CoT and GCA-based traffic signal controllers results in statistically significant improvement across different demand patterns compared to three conventional controllers. On average, the ZS-CoT controller managed to have 2.60% fewer vehicles present at the same time compared to the fixed-time controller; compared to the gap-based and delay-based controllers, the ZS-CoT controller achieved 14.22% and 5.89%, respectively. Delving into the details, it is evident that most of this improvement is attributed to the third pattern, where the conventional controllers could not handle tidal traffic patterns. Even without the third pattern, the ZS-CoT controller improved this metric by 5.82% and 1.16% compared to gap-based and delay-based controllers.

The GCA-based controller surpasses the ZS-CoT controller performance. When benchmarked against the fixed-time controller, the GCA-based variant achieved a notable reduction, with 25.89% fewer vehicles present in the network at the same time. In contrast with the gap-based and delay-based controllers, the GCA-based controller enhanced this specific performance metric by 36.32% and 29.47%, respectively. When excluding the third pattern, these improvements are registered at 33.02% and 33.25%.

These results suggest that integrating an actor-critic (AC) feedback mechanism bolsters the long-term planning capabilities of traffic controllers. Intriguingly, such a trend is manifest across all demand patterns, underscoring the robustness of the GCA-based approach. The GCA-based controller generally demonstrates a more pronounced significance of change.

Across all patterns, the ZS-CoT and the GCA-based controllers strategically prioritized critical approaches, occasionally at the expense of less crucial ones (Figure 7). This strategic allocation is especially pronounced in Pattern 3, where two adjacent approaches are critical, limiting phase design due to high-volume conflicting movements. In Pattern 4, which tests the controllers under dynamic demand conditions, the enhanced performance of the GCA-based controller becomes even more discernible.

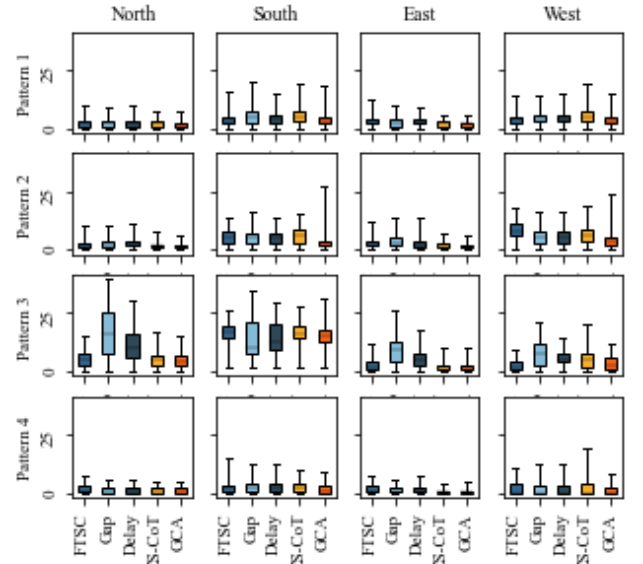


Fig. 7. Total number of vehicles present in the network at the approach level.

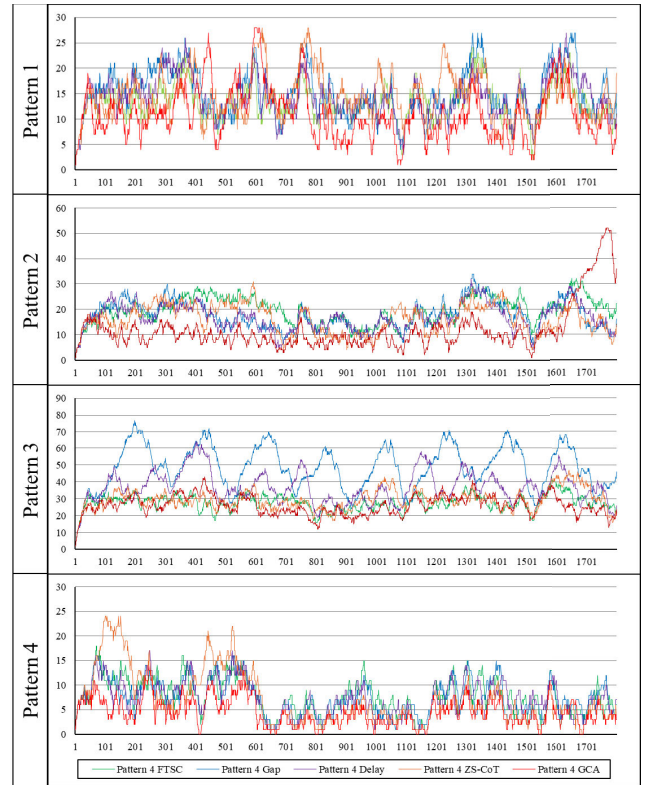


Fig. 8. An example of comparative analysis of controllers on the number of vehicles present in the network as a measure of continuous performance and stability.

The total number of vehicles present in the network at any given time can also serve as an effective stability measure since a stable system effectively manages this balance to prevent overwhelming buildup. If the number of vehicles within the network remains relatively constant over time or grows slowly, it suggests that the controller is maintaining stability despite fluctuations in traffic demand. Figure 8 depicts an example of this performance measure across a 15-minute period. In the majority of instances, LLM-based

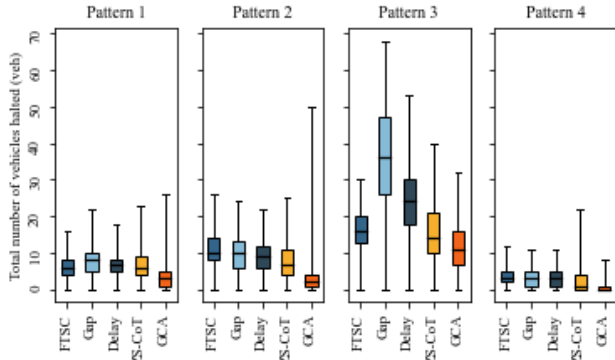


Fig. 9. Total number of vehicles halted in the network at the intersection level.

controllers demonstrate superior performance, evidenced by their heightened stability (evidenced by lower variance) and their consistent success in maintaining fewer vehicles within the network—indicative of enhanced performance. This superiority is particularly noteworthy in the third and fourth traffic patterns, which present more complex challenges. Despite the dynamic nature of traffic demand, the LLM-based controllers consistently maintain performance in the lower variance band. In pattern 4, notably, after the 1200th time step, where a spike in travel demand is introduced, all controller types visibly grapple with the new traffic influx. However, the LLM-based controllers exhibit a notably swifter response to this abrupt change, rapidly stabilizing the intersection.

b) Total number of vehicles halted (HV): A halted vehicle is a vehicle in the network with a speed below 0.1 m/s. The number of vehicles halted at intersections provides an insight into how the traffic signal controller manages to avoid exacerbated travel times.

An increase in this performance measure inevitably led to elevated waiting times. However, the implications extend beyond mere waiting. It intensifies the stop-and-go traffic pattern, increasing the likelihood of rear-end collisions and causing longer queue lengths at intersections. Moreover, such halting can provoke driver frustration, potentially resulting in aggressive driving patterns that compromise the safety of all road users.

LLM-based controllers, especially the GCA-based controller, have demonstrated high proficiency in curtailing the total count of vehicles halted at intersections (Figure 9). Statistical validation via the Wilcoxon signed-rank test, as detailed in TABLE V, confirms this observation, indicating that the deployment of ZS-CoT and GCA-based controllers offers statistically significant advancements over conventional benchmark controllers across various scenarios. Notably, the evaluation metric underscores the frequency of vehicle stops, where a reduced count signifies a more optimal outcome. When benchmarked against conventional controllers, the ZS-CoT controller exhibited reductions in halted vehicles by 8.28%, 25.75%, and 14.01% compared to fixed-time, gap-based, and delay-based controllers, respectively. However, the GCA-based controller outperformed the ZS-CoT in this metric, too. When assessed against the conventional controllers, it showcased reductions of 48.03%, 60.66%, and 53.38%.

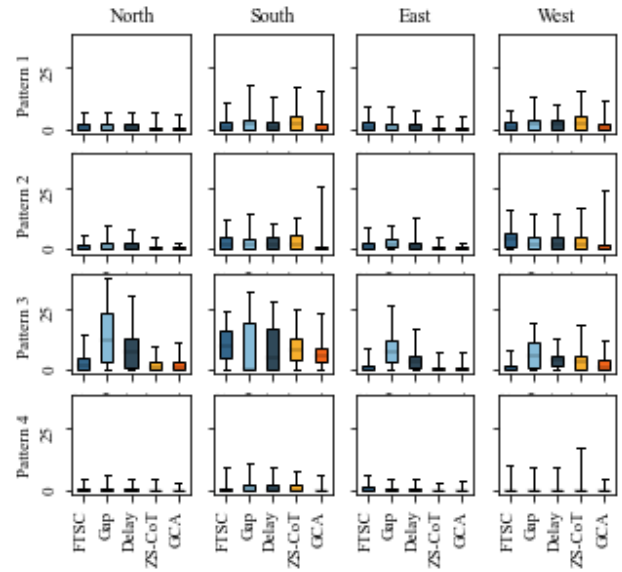


Fig. 10. Total vehicles that are halted at the approach level.

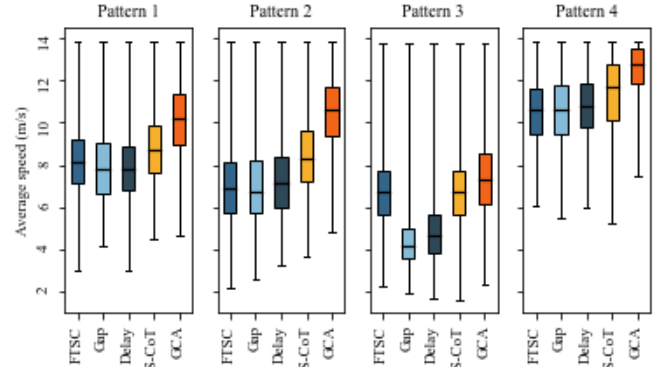


Fig. 11. Vehicles' average speed at the intersection level.

Mirroring previous performance measures at the approach level, the ZS-CoT and GCA-based controllers prioritize critical approaches to boost the overall intersection performance (Figure 10). Notably, as the traffic patterns become more complex or dynamic, the efficacy of both LLM-based controllers becomes increasingly apparent. For instance, in the third pattern, both LLM-based controllers significantly reduced the number of halted vehicles in critical north and east approaches, outperforming other controllers.

c) Average speed (AS): The average speed of the vehicles offers a nuanced insight into the dynamic interplay between traffic flow and signal timing. A well-optimized controller should, ideally, facilitate the smooth passage of vehicles, thereby ensuring a higher average speed.

Conversely, a suboptimal controller may unnecessarily interrupt the traffic flow, causing reductions in average speed. Hence, when evaluating the performance of a traffic signal controller, the average speed emerges as an important metric.

Figure 11 illustrates the average speed at the intersection level. Since the desired speed of the users is set to be 13.42 m/s, any average speed close to this ideal speed can be an indicative measure of a controller's competence. A thorough examination of Figure 11 reveals that LLM-based

TABLE V

INTERSECTION-LEVEL COMPARATIVE ANALYSIS USING LLM-BASED AND CONVENTIONAL CONTROLLERS: THIS TABLE PRESENTS THE RESULTS OF THE WILCOXON SIGNED-RANK TEST, COMPARING THE PERFORMANCE OF INTERSECTIONS CONTROLLED BY LLM-BASED CONTROLLERS VERSUS CONVENTIONAL CONTROLLERS ALONGSIDE THE AVERAGE CHANGES OBSERVED IN PERCENTAGE. NUMBERS HIGHLIGHTED IN GREEN INDICATE THE HIGHEST FAVORABLE CHANGES, WHILE RED HIGHLIGHTS DENOTE INSIGNIFICANT CHANGES BASED ON THE P-VALUES OF THE TEST

PM	Pattern	FTSC				Gap- based				Delay-based			
		ZS-CoT		GCA-based		ZS-CoT		GCA-based		ZS-CoT		GCA-based	
		Change	P-value	Change	P-value	Change	P-value	Change	P-value	Change	P-value	Change	P-value
NVP	P1	8.14	9.3E-19	-17.31	4.1E-114	-5.09	8.6E-16	-27.43	5.0E-174	1.55	4.6E-01	-22.35	3.1E-148
	P2	-17.17	2.8E-138	-39.17	9.3E-184	-7.95	3.1E-33	-32.4	9.3E-184	-3.25	3.1E-07	-28.95	2.1E-157
	P3	5.55	1.1E-34	-6.26	9.3E-53	-39.42	1.0E-288	-46.2	1.6E-291	-20.09	3.6E-175	-29.03	2.1E-281
	P4	-6.93	9.5E-25	-40.84	3.0E-254	-4.42	3.4E-20	-39.24	2.1E-261	-1.77	8.7E-13	-37.56	1.1E-241
HV	P1	12.36	2.9E-07	-35.82	1.9E-109	-12.46	4.1E-23	-50	3.2E-170	0.01	7.5E-02	-42.88	1.6E-136
	P2	-27.83	5.0E-103	-54.37	1.6E-167	-22.2	2.6E-57	-50.81	3.1E-153	-14.92	1.9E-25	-46.21	2.2E-148
	P3	-2.8	5.0E-03	-26.75	5.6E-109	-57.18	1.8E-284	-67.73	1.0E-289	-36.77	1.3E-176	-52.35	6.1E-283
	P4	-14.86	1.1E-29	-75.16	3.3E-234	-11.16	2.6E-27	-74.08	2.8E-242	-4.34	8.9E-15	-72.1	9.1E-219
CD	P1	11.87	2.0E-12	-22.63	7.0E-56	-8.08	2.3E-10	-36.43	3.1E-117	3.07	4.0E-01	-28.71	6.5E-82
	P2	12.61	2.6E-05	-10.06	2.3E-68	-22.93	1.1E-87	-38.44	2.6E-132	-10.39	8.7E-26	-28.43	5.1E-117
	P3	28.18	6.8E-20	-0.15	4.7E-10	-64.99	4.3E-288	-72.73	1.5E-291	-42.68	2.5E-176	-55.35	1.7E-270
	P4	-12.81	1.2E-11	-72.36	6.1E-200	-6.76	2.2E-09	-70.45	6.5E-214	3.97	2.2E-01	-67.05	1.9E-177
AS	P1	7.32	6.5E-55	23.36	2.1E-212	11.43	1.7E-81	28.08	2.8E-221	10.92	2.9E-98	27.49	7.4E-231
	P2	21.7	1.2E-181	49.61	3.1E-280	20.32	2.5E-166	47.92	1.1E-273	15.68	4.0E-130	42.22	2.0E-278
	P3	-0.95	3.9E-01	9.8	2.4E-46	51.18	1.0E-243	67.6	6.7E-289	37.25	5.1E-220	52.16	3.6E-284
	P4	7.71	4.8E-82	18.39	3.6E-258	7.58	9.6E-98	18.25	2.7E-275	5.2	2.0E-49	15.63	1.0E-253

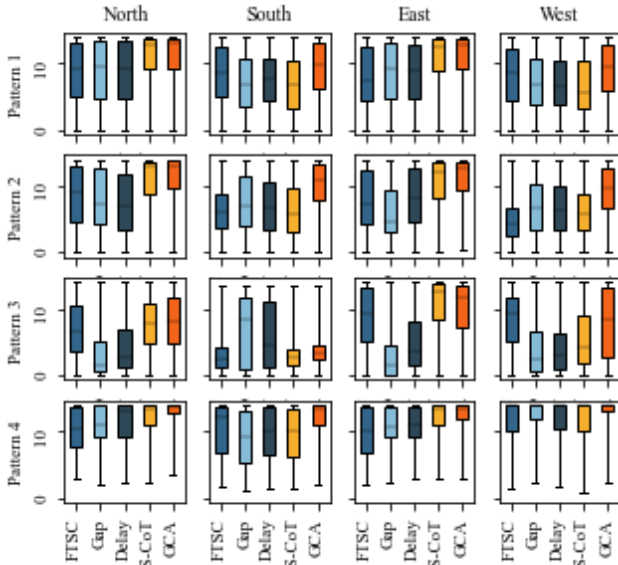


Fig. 12. Vehicles' average speed at the approach level.

controllers consistently exhibit better performance in maintaining average speeds, underscoring their potential efficacy in real-world traffic management scenarios. TABLE V presents a detailed quantitative comparison of the average speed for traffic demand patterns and controller combinations derived from the Wilcoxon signed-rank test. These results reveal that aside from Pattern 3—where the ZS-CoT controller's performance relative to the fixed-time controller was statistically insignificantly less efficient—the LLM-based controllers consistently outperformed their conventional counterparts, the fixed-time, gap-based, and delay-based controllers. The ZS-CoT controller enhanced the average flow speed by 8.94%, 22.63%, and 17.26%, respectively. Meanwhile, the GCA-based controller further advanced on these gains, registering improvements of 25.29%, 40.46%, and 34.37% compared to the conventional controllers.

An in-depth analysis of the average speed at the approach level underscores the propensity of both LLM-based controllers to prioritize critical approaches (Figure 12). Specifically, the ZS-CoT controller manifested improvements in average speed for critical approaches at 16.86%, 55.94%, and 30.75% over the fixed-time, gap-based, and delay-based controllers, respectively.

In contrast, the improvements were modest for non-critical approaches: 1.28%, 4.73%, and 4.76% compared with the corresponding controllers. The differences become even more pronounced when evaluating the GCA-based controller. In critical approaches, the GCA-based controller amplified the average speed by 30.26%, 67.81%, and 42.71% relative to the fixed-time, gap-based, and delay-based controllers. Even in the non-critical approaches, the gains were noteworthy: 23.59%, 28.64%, and 28.79% improvements over the same conventional benchmarks. Notably, even non-critical approaches controlled by the GCA-based controller had a 6.37% higher average speed than the critical approaches controlled by the ZS-CoT controller. This sheer contrast underscores the higher efficacy of the GCA-based compared to the ZS-CoT.

3) *Traffic Controller-Specific Performance Measure:* In addition to the traffic flow characteristics, the performance of the controller can also be measured by the extent of disruption they cause to traffic, which is quantified using control delay and the percentage of arrivals on the green.

a) *Control delay (CD):* The control delay can characterize the controllers' performance at the intersection level and is one of the main criteria for determining the level of service (LOS) at signalized intersections [75]. Given that all the scenarios considered in this study are undersaturated, the control delay is the sole factor influencing the intersection's level of service. Control delay provides a way to measure the congestion and the inefficiencies in signal timings directly. By focusing on the waiting time of individual vehicles, it captures not only the collective users' experience but also reveals the granular experiences of singular users.

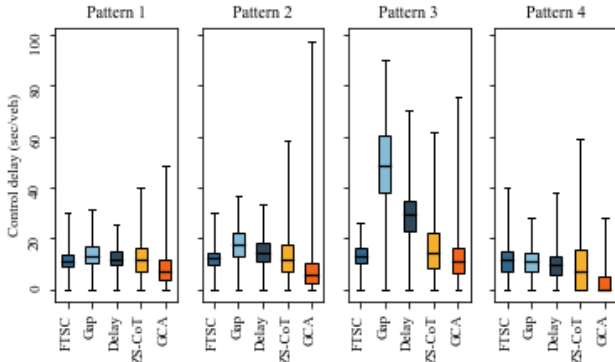


Fig. 13. Control delay at the intersection level.

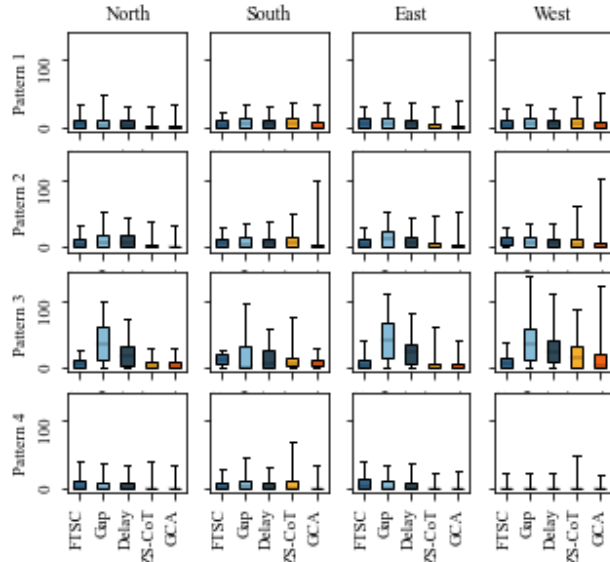


Fig. 14. Control delay at the approach level.

Figure 13 illustrates the variations of these performance measures at the intersection level, where the LLM-based controllers, especially the GCA-based controller, demonstrated a significant reduction in the average waiting time each vehicle experienced. TABLE V provides more quantitative insight into the outcomes of this performance measure at the intersection level based on the Wilcoxon signed-rank test. As is evident from this table, the ZS-CoT controller failed to improve the average waiting time per vehicle compared to the fixed-time controller, increasing the waiting time by 9.96%. However, compared to gap-based and delay-based controllers, the ZS-CoT controller decreased the average waiting time by 25.96% and 11.51%, respectively.

Conversely, the GCA-based controller outperformed all the patterns and benchmarks. Compared to fixed-time, gap-based, and delay-based controllers, the GCA-based controller reduced the control delay by 36.30%, 54.51%, and 44.89%. This significant improvement in this important performance measure directly connected to the interaction structure shows how text-based knowledge can improve the actor agent's performance.

At the approach level (Figure 14), distinct variations in controllers' efficiency are evident across traffic patterns. Using the GCA-based approach resulted in an improve-

TABLE VI
PERCENTAGE OF ARRIVAL ON GREEN (POG)

Controller	Pattern 1	Pattern 2	Pattern 3	Pattern 4
FTSC	53.72%	45.25%	42.40%	52.10%
Gap-based	47.87%	43.55%	24.95%	52.86%
Delay-based	51.18%	45.75%	32.96%	55.00%
ZS-CoT	51.92%	52.29%	46.95%	56.18%
GCA-based	64.09%	58.93%	54.99%	79.89%

ment in performance across all directions for all patterns. The results from the Wilcoxon signed-rank test reveal some contrasting trends for the ZS-CoT controller compared to the fixed time controller. Specifically, in critical approaches, the ZS-CoT controller managed to reduce the waiting time by 17.12%. However, in other approaches, this controller exhibited an increase in waiting time by 19.94%. This reveals that the ZS-CoT was willing to sacrifice its performance with non-critical approaches to improve its overall performance. Compared to the gap-based and delay-based controllers, the ZS-CoT controllers improved the waiting time in both critical approaches by 36.75% and 30.70%, respectively, and non-critical approaches by 26.04% and 11.99%, respectively.

On the other hand, the GCA-based controller improved the average waiting time that each vehicle experiences compared to other scenarios controlled by conventional controllers. In critical approaches, the GCA-based controllers improved the average waiting time per vehicle by 44.80%, 64.52%, and 58.56% compared to conventional controllers, fixed-time, gap-based, and delay-based controllers, respectively. In non-critical approaches, the GCA-based controller showcased improvements in the average waiting time per vehicle by margins of 31.59%, 59.21%, and 53.01%, respectively. Across all pattern-controller combinations, only in one scenario, shifting from the fixed time controller to the GCA-based controller within pattern 1, the GCA-based controller sacrificed the performance in the west approach, a non-critical approach, in favor of the overall performance.

b) Percentage of Arrival on Green (POG): POG is generally considered an intersection-level performance measure [76], which is one of the more nuanced performance measures for measuring the performance of traffic signals. This metric quantifies the proportion of vehicles that encounter a green signal upon arriving at an intersection.

A higher ratio suggests a controller's proficiency in synchronizing traffic flows with signal timings, reducing unnecessary stops and delays. Consequently, adopting POG as a performance measure can provide invaluable insights into the real-time operational efficiency of traffic controllers, serving as a reliable metric to gauge their adaptability and responsiveness.

As provided in TABLE VI, which depicts this performance measure for each controller, implementing the LLM-based controllers, especially the GCA-based controller, increases the percentage of vehicles that arrive on green time. For instance, in the fourth pattern, which provides a dynamic demand pattern, the GCA-based controller managed to increase this performance measure from 52.10%, 52.86%, and 55.00% for fixed-time, gap-based, and delay-based controllers to 79.89%. This significant increase reveals the superiority of

TABLE VII

PERCENTAGE OF TIMES THE SAFETY MECHANISM WAS TRIGGERED

Controller	Pattern 1	Pattern 2	Pattern 3	Pattern 4
ZS-CoT	1.48%	7.36%	8.84%	2.20%
GCA-based	2.20%	0.00%	2.96%	1.48%

the LLM-based controllers in avoiding avoidable delays by strategically managing the green lights.

4) *Algorithm Safety Performance Measure*: As discussed before, a safety mechanism was designed to avoid unsafe or unfeasible strategies. This mechanism ensures that the generated phase considers the safety of conflicting movement.

Since the LLM-based controllers can devise new traffic light patterns, the ratio of the times this mechanism is triggered at decision points serves as a straightforward indicator of the reliability of the reasoning and planning mechanisms. For instance, in Pattern 3, where the major approaches are adjacent (west and east), allowing both approaches through movement to use the intersection simultaneously would be dangerous, creating an unsafe phase. The percentage of times that the safety mechanism takes over and intervenes in the devised phase plans or durations was considered to evaluate the reasoning and planning process's safety. TABLE VII provides the percentage of times the safety mechanism was triggered throughout this study by giving a ratio of the times this mechanism was triggered at decision points.

In this study, controllers are exposed to more complex traffic scenarios in each traffic pattern. A noteworthy observation is the apparent relationship between the complexity of these traffic scenarios and the increased likelihood of safety mechanisms being triggered. However, when comparing the two controllers within the same traffic patterns, implementing the text-based knowledge system can be an effective strategy that can help reduce the chance of failure of the reasoning and planning process. This highlights the importance of implementing a safety mechanism when developing similar agent-based controllers that involve reasoning and planning.

VI. CONCLUSION

This study demonstrated the potential of LLM-based approaches in the development of adaptive traffic signal controllers. Within this context, a comparative analysis of two constructed LLM-based controllers, Zero-Shot Chain of Thought (ZS-CoT) and Generally Capable Agent (GCA)-based controllers, with three conventional benchmarks, fixed-time, gap-based, and delay-based signal control systems, was conducted using SUMO. Specifically, during this study, we evaluated the developed LLM-based controllers in four dimensions: (1) the ability of the controllers to adapt to traffic patterns and devise new traffic patterns or adjust phase duration, (2) the ability to improve traffic flow performance measures, (3) the ability to improve traffic controller-specific performance measure, and (4) algorithm safety performance measure. The developed models were tested in challenging scenarios, including symmetrical, asymmetrical, tidal, and variable traffic demand patterns. The results are presented at intersection and approach levels to set out a comprehensive overview of the results and to facilitate a deeper understanding of how the LLM-based controller managed the critical

movements. The Wilcoxon signed-rank test was used at each level to compare the controllers' performance measures.

Results at both intersection and approach levels indicated that LLM-based controllers excelled by creating more diverse signal phases, reducing phase changes, and optimizing cycle time, leading to reduced vehicle numbers and delays, improved traffic speeds, and increased arrival rates on the green. The GCA-based controllers, with their actor-critic feedback mechanism, outperformed ZS-CoT controllers by refining strategies in real-time and significantly reducing unsafe phase patterns, highlighting their advanced reasoning and planning capabilities in traffic management.

This study highlights the effectiveness of LLM-based controllers in managing a traffic-related reasoning and planning task. It shows that while the zero-shot approach relying solely on an LLM's pre-existing knowledge is feasible for complex tasks like managing signalized intersections, enhancements are still achievable through online knowledge extraction. Furthermore, the incorporation of a secondary "critic agent" to refine this knowledge can further improve the controllers' performance. Such integration allows the formation of a closed-loop controller that markedly boosts performance by enhancing the system's reasoning capabilities and action planning. The findings demonstrate the impressive ability of LLM-based traffic controllers to adeptly manage the complexities of dynamic, real-time traffic environments.

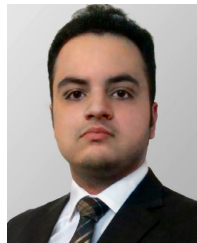
While this study demonstrates the potential of using LLM-based controllers, it bears several limitations. First, it is limited to one isolated intersection in hypothetical demand scenarios. To comprehend these controllers' true efficiency and their ability to cooperate and coordinate with other controllers, evaluating their performance within a real-world network of intersections is imperative. Moreover, the study did not consider other modes of transportation, such as emergency vehicles and buses, thus limiting its applicability in more versatile urban scenarios. Furthermore, the potential alterations in traffic flow due to external factors, such as weather-related events and traffic incidents, remained unexplored. These external factors possess the potential to impact the traffic flow characteristics, thereby influencing the traffic signal controllers' strategy. Finally, the study's performance evaluation was limited to a few performance measures. Future research should consider a broader range of performance measures to provide a more comprehensive assessment of the controllers' effectiveness. The application of LLM-based controllers could also be extended to other areas facing challenging and complex environments, including asset management [77], [78], [79], incident management [80], and evacuation planning [81]. Such studies would address the scalability of LLM-based controllers and their effectiveness in real-world environments that reflect the complexity of urban traffic systems. Advancing this research could solidify the foundational understanding established by this study, moving towards more practical and robust traffic management solutions.

REFERENCES

- [1] S. Bubeck et al., "Sparks of artificial general intelligence: Early experiments with GPT-4," 2023, *arXiv:2303.12712*.
- [2] L. Xie et al., "Towards AGI in computer vision: Lessons learned from GPT and large language models," 2023, *arXiv:2306.08641*.

- [3] C. Zhang et al., "One small step for generative AI, one giant leap for AGI: A complete survey on ChatGPT in AIGC era," 2023, *arXiv:2304.06488*.
- [4] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, Jun. 2017, pp. 5999–6009.
- [5] S. Hao et al., "Reasoning with language model is planning with world model," 2023, *arXiv:2305.14992*.
- [6] Z. Wang, S. Cai, G. Chen, A. Liu, X. Ma, and Y. Liang, "Describe, explain, plan and select: Interactive planning with large language models enables open-world multi-task agents," 2023, *arXiv:2302.01560*.
- [7] L. Wang et al., "Plan-and-solve prompting: Improving zero-shot chain-of-thought reasoning by large language models," 2023, *arXiv:2305.04091*.
- [8] INRIX. (Feb. 2018). *INRIX Global Traffic Scorecard*. [Online]. Available: <https://media.bizj.us/view/img/10360454/inrix2016trafficscorecarden.pdf>
- [9] R. Schuman. (Apr. 2022). *INRIX US Signals Scorecard*. INRIX. [Online]. Available: <https://inrix.com/signals-scorecard/>
- [10] Y. Wang, X. Yang, H. Liang, and Y. Liu, "A review of the self-adaptive traffic signal control system based on future traffic environment," *J. Adv. Transp.*, vol. 2018, pp. 1–12, Jun. 2018, doi: [10.1155/2018/1096123](https://doi.org/10.1155/2018/1096123).
- [11] M. Guo, P. Wang, C.-Y. Chan, and S. Askary, "A reinforcement learning approach for intelligent traffic signal control at urban intersections," in *Proc. IEEE Intell. Transp. Syst. Conf. (ITSC)*, Oct. 2019, pp. 4242–4247, doi: [10.1109/ITSC.2019.8917268](https://doi.org/10.1109/ITSC.2019.8917268).
- [12] T. Urbanik et al., *Signal Timing Manual*, 2nd ed., Washington, DC, USA: Transportation Research Board, Sep. 2015, doi: [10.17226/22097](https://doi.org/10.17226/22097).
- [13] P. B. Hunt, D. I. Robertson, R. D. Bretherton, and R. I. Winton, "SCOOT—A traffic responsive method of coordinating signals," TRRL Lab. Rep. (Transport Road Res. Lab., Great Britain), Wokingham, U.K., Tech. Rep. LR 1014 Monograph, 1981. [Online]. Available: <https://trid.trb.org/View/179439>
- [14] A. G. Sims and K. W. Dobinson, "The Sydney coordinated adaptive traffic (SCAT) system philosophy and benefits," *IEEE Trans. Veh. Technol.*, vol. VT-29, no. 2, pp. 130–137, May 1980, doi: [10.1109/TVT.1980.23833](https://doi.org/10.1109/TVT.1980.23833).
- [15] A. Ansariyar and M. Jeihani, "Real-time traffic control and safety measure analysis using LiDAR sensor during traffic signal failures," in *Proc. Int. Conf. Transp. Develop.*, Jun. 2024, pp. 798–810, doi: [10.1061/9780784485514.070](https://doi.org/10.1061/9780784485514.070).
- [16] T. Ardalan, D. Sarazhinsky, N. Dobrota, S. Gavric, and A. Stevanovic, "Overview and modeling capabilities of an event-based signal controller," *Symmetry*, vol. 16, no. 2, p. 157, Jan. 2024, doi: [10.3390/sym16020157](https://doi.org/10.3390/sym16020157).
- [17] N. Choosri, "IoT-RFID testbed for supporting traffic light control," *Int. J. Inf. Electron. Eng.*, vol. 5, no. 2, p. 102, Mar. 2015, doi: [10.7763/ijee.2015.v5.511](https://doi.org/10.7763/ijee.2015.v5.511).
- [18] S. Damadam, M. Zourbakhsh, R. Javidan, and A. Faroughi, "An intelligent IoT based traffic light management system: Deep reinforcement learning," *Smart Cities*, vol. 5, no. 4, pp. 1293–1311, Sep. 2022, doi: [10.3390/smartcities5040066](https://doi.org/10.3390/smartcities5040066).
- [19] S. A. Kashinath et al., "Review of data fusion methods for real-time and multi-sensor traffic flow analysis," *IEEE Access*, vol. 9, pp. 51258–51276, 2021, doi: [10.1109/ACCESS.2021.3069770](https://doi.org/10.1109/ACCESS.2021.3069770).
- [20] H. Wei, G. Kashyap, D. Nian, and W. Lin, "V2I-based intelligent mobility data fusion for self-adaptive traffic signal control," in *Proc. Int. Conf. Transp. Develop.*, vol. 2, Aug. 2022, pp. 200–211, doi: [10.1061/9780784484326.020](https://doi.org/10.1061/9780784484326.020).
- [21] T. Anagnostopoulos, D. Ferreira, A. Samodelkin, M. Ahmed, and V. Kostakos, "Cyclist-aware traffic lights through distributed smartphone sensing," *Pervas. Mobile Comput.*, vol. 31, pp. 22–36, Sep. 2016, doi: [10.1016/j.pmcj.2016.01.012](https://doi.org/10.1016/j.pmcj.2016.01.012).
- [22] H. Maripini, L. Vanajakshi, and B. R. Chilukuri, "A probe-based demand responsive signal control for isolated intersections under mixed traffic conditions," *Transp. Lett.*, vol. 16, no. 2, pp. 117–130, Feb. 2024, doi: [10.1080/19427867.2022.2164613](https://doi.org/10.1080/19427867.2022.2164613).
- [23] S. Mishra, V. Singh, A. Gupta, D. Bhattacharya, and A. Mudgal, "Adaptive traffic signal control for developing countries using fused parameters derived from crowd-source data," *Transp. Lett.*, vol. 15, no. 4, pp. 296–307, Apr. 2023, doi: [10.1080/19427867.2022.2050493](https://doi.org/10.1080/19427867.2022.2050493).
- [24] K. K. Sharma and S. Indu, "GPS based adaptive traffic light timings and lane scheduling," in *Proc. IEEE Intell. Transp. Syst. Conf. (ITSC)*, Oct. 2019, pp. 4267–4274, doi: [10.1109/ITSC.2019.8917153](https://doi.org/10.1109/ITSC.2019.8917153).
- [25] R. Tanwar, R. Majumdar, G. S. Sidhu, and A. Srivastava, "Removing traffic congestion at traffic lights using GPS technology," in *Proc. 6th Int. Conf. Cloud Syst. Big Data Eng.*, Jan. 2016, pp. 575–579, doi: [10.1109/CONFLUENCE.2016.7508185](https://doi.org/10.1109/CONFLUENCE.2016.7508185).
- [26] T. Yoshioka, H. Sakakibara, R. Tenhagen, S. Lorkowski, and T. Oguchi, "Traffic signal control parameter calculation using probe data," *Int. J. Intell. Transp. Syst. Res.*, vol. 20, no. 1, pp. 288–298, Apr. 2022, doi: [10.1007/s13177-021-00292-z](https://doi.org/10.1007/s13177-021-00292-z).
- [27] N. J. Goodall, B. L. Smith, and B. Park, "Traffic signal control with connected vehicles," *Transp. Res. Rec., J. Transp. Res. Board*, vol. 2381, no. 1, pp. 65–72, Jan. 2013, doi: [10.3141/2381-08](https://doi.org/10.3141/2381-08).
- [28] A. Qu, Y. Tang, and W. Ma, "Adversarial attacks on deep reinforcement learning-based traffic signal control systems with colluding vehicles," *ACM Trans. Intell. Syst. Technol.*, vol. 14, no. 6, pp. 1–22, Nov. 2023, doi: [10.1145/3625236](https://doi.org/10.1145/3625236).
- [29] C. Cai, Y. Wang, and G. Geers, "Vehicle-to-infrastructure communication-based adaptive traffic signal control," *IET Intell. Transp. Syst.*, vol. 7, no. 3, pp. 351–360, Sep. 2013, doi: [10.1049/iet-its.2011.0150](https://doi.org/10.1049/iet-its.2011.0150).
- [30] A. Jovanović and D. Teodorović, "Fixed-time traffic control at super-street intersections by bee colony optimization," *Transp. Res. Rec., J. Transp. Res. Board*, vol. 2676, no. 4, pp. 228–241, Apr. 2022, doi: [10.1177/03611981211058104](https://doi.org/10.1177/03611981211058104).
- [31] C. Louw, L. Labuschagne, and T. Woodley, "A comparison of reinforcement learning agents applied to traffic signal optimisation," in *Proc. SUMO Conf.*, vol. 3, Sep. 2022, pp. 15–43, doi: [10.52825/sep.v3i.116](https://doi.org/10.52825/sep.v3i.116).
- [32] S. G. Shelby, "Single-intersection evaluation of real-time adaptive traffic signal control algorithms," *Transp. Res. Rec., J. Transp. Res. Board*, vol. 1867, no. 1, pp. 183–192, Jan. 2004, doi: [10.3141/1867-21](https://doi.org/10.3141/1867-21).
- [33] M. Yazdani, M. Sarvi, S. A. Bagloee, N. Nassir, J. Price, and H. Parineh, "Intelligent vehicle pedestrian light (IVPL): A deep reinforcement learning approach for traffic signal control," *Transp. Res. C, Emerg. Technol.*, vol. 149, Apr. 2023, Art. no. 103991, doi: [10.1016/j.trc.2022.103991](https://doi.org/10.1016/j.trc.2022.103991).
- [34] C. Day et al., "Performance measures for traffic signal systems: An outcome-oriented approach," Purdue Univ., West Lafayette, IN, USA, Tech. Rep., Apr. 2014, doi: [10.5703/1288284315333](https://doi.org/10.5703/1288284315333). [Online]. Available: <http://docs.lib.purdue.edu/jtrpaffdocs/3/>
- [35] J. Achiam et al., "GPT-4 technical report," 2023, *arXiv:2303.08774*.
- [36] A. Chowdhery et al., "PaLM: Scaling language modeling with pathways," 2022, *arXiv:2204.02311*.
- [37] D. Wang, Y. Fu, K. Liu, F. Chen, P. Wang, and C.-T. Lu, "Automated urban planning for reimagining city configuration via adversarial learning: Quantification, generation, and evaluation," *ACM Trans. Spatial Algorithms Syst.*, vol. 9, no. 1, pp. 1–24, Mar. 2023, doi: [10.1145/3524302](https://doi.org/10.1145/3524302).
- [38] P. J. Giabbanelli, "GPT-based models meet simulation: How to efficiently use large-scale pre-trained language models across simulation tasks," 2023, *arXiv:2306.13679*.
- [39] F. Guo, "GPT in game theory experiments," 2023, *arXiv:2305.05516*.
- [40] J. Sung Park, J. C. O'Brien, C. J. Cai, M. R. Morris, P. Liang, and M. S. Bernstein, "Generative agents: Interactive simulacra of human behavior," 2023, *arXiv:2304.03442*.
- [41] H. Yang, S. Yue, and Y. He, "Auto-GPT for online decision making: Benchmarks and additional opinions," 2023, *arXiv:2306.02224*.
- [42] T. Kojima, S. Shane Gu, M. Reid, Y. Matsuo, and Y. Iwasawa, "Large language models are zero-shot reasoners," 2022, *arXiv:2205.11916*.
- [43] J. Wei et al., "Chain-of-thought prompting elicits reasoning in large language models," 2022, *arXiv:2201.11903*.
- [44] X. Zhu et al., "Ghost in the minecraft: Generally capable agents for open-world environments via large language models with text-based knowledge and memory," 2023, *arXiv:2305.17144*.
- [45] Y. Wu et al., "SPRING: Studying the paper and reasoning to play games," 2023, *arXiv:2305.15486*.
- [46] M. Ahn et al., "Do as I can, not as I say: Grounding language in robotic affordances," 2022, *arXiv:2204.01691*.
- [47] I. Dasgupta et al., "Collaborating with language models for embodied reasoning," 2023, *arXiv:2302.00763*.
- [48] J. L. Espejel, E. H. Ettifouri, M. S. Y. Alassan, E. M. Chouham, and W. Dahhane, "GPT-3.5, GPT-4, or BARD? Evaluating LLMs reasoning ability in zero-shot setting and performance boosting through prompts," 2023, *arXiv:2305.12477*.
- [49] M. Han, F. Chen, J. Shi, S. Xu, and B. Xu, "Knowledge transfer from pre-trained language models to cif-based speech recognizers via hierarchical distillation," 2023, *arXiv:2301.13003*.
- [50] A. J. Miller, "A computer control system for traffic networks," in *Proc. 2nd Int. Symp. Theory Traffic Flow*, 1963, pp. 200–220.
- [51] M. Nakatsuyama, H. Nagahashi, and N. Nishizuka, "Fuzzy logic phase controller for traffic junctions in the one-way arterial road," *IFAC Proc. Volumes*, vol. 17, no. 2, pp. 2865–2870, Jul. 1984, doi: [10.1016/s1474-6670\(17\)61417-4](https://doi.org/10.1016/s1474-6670(17)61417-4).

- [52] C. P. Pappis and E. H. Mamdani, "A fuzzy logic controller for a traffic junction," *IEEE Trans. Syst., Man, Cybern.*, vol. SMC-7, no. 10, pp. 707–717, Oct. 1977, doi: [10.1109/TSMC.1977.4309605](https://doi.org/10.1109/TSMC.1977.4309605).
- [53] R. Saeidi-Razavi, P. G. Furth, and M. Tahmasebi, "Transit-friendly underlying traffic signal control, the key to effective transit signal priority," in *Proc. Transp. Res. Board Meeting*, Washington, DC, USA, 2022. [Online]. Available: <https://bpb-us-e1.wpmucdn.com/sites.northeastern.edu/dist/e/618/files/2023/02/10-Transit-Friendly-Signal-Control.pdf>
- [54] J. Liu et al., "Traffic signal control using reinforcement learning based on the teacher–student framework," *Exp. Syst. Appl.*, vol. 228, Oct. 2023, Art. no. 120458, doi: [10.1016/j.eswa.2023.120458](https://doi.org/10.1016/j.eswa.2023.120458).
- [55] S. El-Tantawy, B. Abdulhai, and H. Abdelgawad, "Design of reinforcement learning parameters for seamless application of adaptive traffic signal control," *J. Intell. Transp. Syst.*, vol. 18, no. 3, pp. 227–245, Jul. 2014, doi: [10.1080/15472450.2013.810991](https://doi.org/10.1080/15472450.2013.810991).
- [56] K. J. Prabhachandran, A. N. H. Kumar, and S. Bhatnagar, "Multi-agent reinforcement learning for traffic signal control," in *Proc. 17th IEEE Int. Conf. Intell. Transp. Syst.*, Nov. 2014, pp. 2529–2534, doi: [10.1109/ITSC.2014.6958095](https://doi.org/10.1109/ITSC.2014.6958095).
- [57] X. Liang, X. Du, G. Wang, and Z. Han, "A deep reinforcement learning network for traffic light cycle control," *IEEE Trans. Veh. Technol.*, vol. 68, no. 2, pp. 1243–1253, Feb. 2019, doi: [10.1109/TVT.2018.2890726](https://doi.org/10.1109/TVT.2018.2890726).
- [58] L. Li, Y. Lv, and F.-Y. Wang, "Traffic signal timing via deep reinforcement learning," *IEEE/CAA J. Autom. Sinica*, vol. 3, no. 3, pp. 247–254, Apr. 2016, doi: [10.1109/JAS.2016.7508798](https://doi.org/10.1109/JAS.2016.7508798).
- [59] S. M. A. Shabestary and B. Abdulhai, "Deep learning vs. discrete reinforcement learning for adaptive traffic signal control," in *Proc. 21st Int. Conf. Intell. Transp. Syst. (ITSC)*, Nov. 2018, pp. 286–293, doi: [10.1109/ITSC.2018.8569549](https://doi.org/10.1109/ITSC.2018.8569549).
- [60] J. Zeng, J. Hu, and Y. Zhang, "Adaptive traffic signal control with deep recurrent Q-learning," in *Proc. IEEE Intell. Vehicles Symp.*, Oct. 2018, pp. 1215–1220, doi: [10.1109/IVS.2018.8500414](https://doi.org/10.1109/IVS.2018.8500414).
- [61] M. Aslani, M. S. Mesgari, and M. Wiering, "Adaptive traffic signal control with actor-critic methods in a real-world traffic network with different traffic disruption events," *Transp. Res. C, Emerg. Technol.*, vol. 85, pp. 732–752, Dec. 2017, doi: [10.1016/j.trc.2017.09.020](https://doi.org/10.1016/j.trc.2017.09.020).
- [62] W. Genders and S. Razavi, "Using a deep reinforcement learning agent for traffic signal control," 2016, *arXiv:1611.01142*.
- [63] E. Javanmardi and S. Liu, "Exploring the human cognitive capacity in understanding systems: A grey systems theory perspective," *Found. Sci.*, vol. 25, no. 3, pp. 803–825, Sep. 2020.
- [64] B. Hu et al., "Enabling intelligent interactions between an agent and an LLM: A reinforcement learning approach," 2023, *arXiv:2306.03604*.
- [65] N. Nascimento, P. Alencar, and D. Cowan, "Self-adaptive large language model (LLM)-based multiagent systems," 2023, *arXiv:2307.06187*.
- [66] P. A. Lopez et al., "Microscopic traffic simulation using SUMO," in *Proc. IEEE Conf. Intell. Transp. Syst.*, Nov. 2018, pp. 2575–2582, doi: [10.1109/ITSC.2018.8569938](https://doi.org/10.1109/ITSC.2018.8569938).
- [67] A. Wegener, M. Piórkowski, M. Raya, H. Hellbrück, S. Fischer, and J.-P. Hubaux, "TraCI: an interface for coupling road traffic and network simulators," in *Proc. 11th Commun. Netw. Simul. Symp.* New York, NY, USA: Association for Computing Machinery, 2008, pp. 155–163, doi: [10.1145/1400713.1400740](https://doi.org/10.1145/1400713.1400740).
- [68] OpenAI. (2024). *GPT Release Notes—OpenAI API*. [Online]. Available: <https://platform.openai.com/docs/gpts/releasenotes>
- [69] L. Miao and D. Leitner, "Adaptive traffic light control with quality-of-service provisioning for connected and automated vehicles at isolated intersections," *IEEE Access*, vol. 9, pp. 39897–39909, 2021, doi: [10.1109/ACCESS.2021.3064310](https://doi.org/10.1109/ACCESS.2021.3064310).
- [70] R. Zhu, W. Ding, S. Wu, L. Li, P. Lv, and M. Xu, "Auto-learning communication reinforcement learning for multi-intersection traffic light control," *Knowl.-Based Syst.*, vol. 275, Sep. 2023, Art. no. 110696, doi: [10.1016/j.knsys.2023.110696](https://doi.org/10.1016/j.knsys.2023.110696).
- [71] F. V. Webster, "Traffic signal settings," Road Res. Lab., London, U.K., Tech. Rep., 39, 1958, p. 44.
- [72] S. Krauß, "Towards a unified view of microscopic traffic flow theories," *IFAC Proc. Volumes*, vol. 30, no. 8, pp. 901–905, Jun. 1997.
- [73] P. Dalggaard, *Introductory Statistics With R* (Statistics and Computing), 2nd ed., New York, NY, USA: Springer, 2008, doi: [10.1007/978-0-387-79054-1](https://doi.org/10.1007/978-0-387-79054-1).
- [74] R. Dowling, "Traffic analysis toolbox volume VI: Definition, interpretation, and calculation of traffic analysis tools measures of effectiveness," Federal Highway Admin., Washington, DC, USA, Tech. Rep. FHWA-HOP-08-054, Jan. 2007, doi: [10.21949/1503647](https://doi.org/10.21949/1503647).
- [75] L. A. Elefteriadou, "The highway capacity manual 6th edition: A guide for multimodal mobility analysis," *ITE J.*, vol. 86, no. 4, pp. 31–99, 2016.
- [76] D. Gettman, E. Fok, E. Curtis, K. C. D. Ormand, M. Mayer, and E. Flanagan, "Measures of effectiveness and validation guidance for adaptive signal control technologies," Federal Highway Admin., Washington, DC, USA, Tech. Rep. FHWA-HOP-13-031, Jul. 2013. [Online]. Available: <https://rosap.nhtl.bts.gov/view/dot/41403>
- [77] W. Hill, H. Jalloul, M. Movahedi, and J. Choi, "Sustainable management of the built environment from the life cycle perspective," *J. Manage. Eng.*, vol. 39, no. 2, Mar. 2023, Art. no. 03123001, doi: [10.1061/JMENEA.MEENG-4759](https://doi.org/10.1061/JMENEA.MEENG-4759).
- [78] A. Bakhshizadeh and K. Sadeghi, "Health-monitoring methods for long-span cable-stayed bridges," *Infrastruct. Asset Manage.*, vol. 11, no. 1, pp. 41–54, Mar. 2024, doi: [10.1680/jinam.23.00030](https://doi.org/10.1680/jinam.23.00030).
- [79] M. Movahedi, R. Khayamim, J. Choi, and E. E. Ozguven, "Network robustness analysis of electric vehicle charging infrastructure during extreme weather conditions: A case study of hurricane ian," in *Proc. Int. Conf. Transp. Develop.* Reston, VA, USA: American Society of Civil Engineers, Jun. 2024, pp. 657–669, doi: [10.1061/9780784485538.058](https://doi.org/10.1061/9780784485538.058).
- [80] A. Grigorev, A.-S. M. K. Saleh, and Y. Ou, "IncidentResponseGPT: Generating traffic incident response plans with generative artificial intelligence," 2024, *arXiv:2404.18550*.
- [81] A. Rafe and P. A. Singleton, "Exploring the complexity of pedestrian dynamics: Impact of societal behaviors and personal attributes in urban environments," *Transp. Res. Rec., J. Transp. Res. Board*, 2024, doi: [10.1177/03611981241260707](https://doi.org/10.1177/03611981241260707).



Mohammad Movahedi received the B.Sc. degree in civil engineering from Yazd University, Yazd, Iran, in 2017, and the M.Sc. degree in transportation engineering from Iran University of Science and Technology, Tehran, Iran, in 2020. He is currently pursuing the Ph.D. degree with the FAMU-FSU College of Engineering. His research interests include decision-making under uncertainty, sustainable infrastructure management, and transportation system analysis.



Juyeong Choi received the B.S. degree in architectural engineering from the University of Seoul, Seoul, South Korea, in 2013, and the M.S. and Ph.D. degrees in civil engineering from Purdue University, West Lafayette, IN, USA, in 2015 and 2018, respectively. Since August 2018, he has been an Assistant Professor with the Department of Civil and Environmental Engineering, FAMU-FSU College of Engineering. His research interests include analysis and modeling of infrastructure system-of-systems and coupled systems (physical-social-ecological)

within the context of sustainability and community resilience. From December 2018 to May 2020, he served as a Co-Guest Editor for a Special Issue on "Management of Resilience in Civil Infrastructure Systems: An Interdisciplinary Approach" of the *Journal of Management in Engineering* (ASCE). He was selected by *Journal of Management in Engineering* for the Runner Up for the Best Peer Reviewed Paper Awards in 2016 and 2020 and the Outstanding Reviewer Award in 2019, 2020, and 2021.